



Inspiring Excellence

CSE370 : Database Systems

Project Report

Project Title : Credit Transfer System

Group No : 02, CSE370 Lab Section : 10, Spring 2026		
ID	Name	Contribution
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24101393	Arunima Lodh	EER Diagram, Normalized Schema, Frontend, Backend
24101404	Aditty Dey	EER Diagram, Normalized Schema, Frontend, Backend

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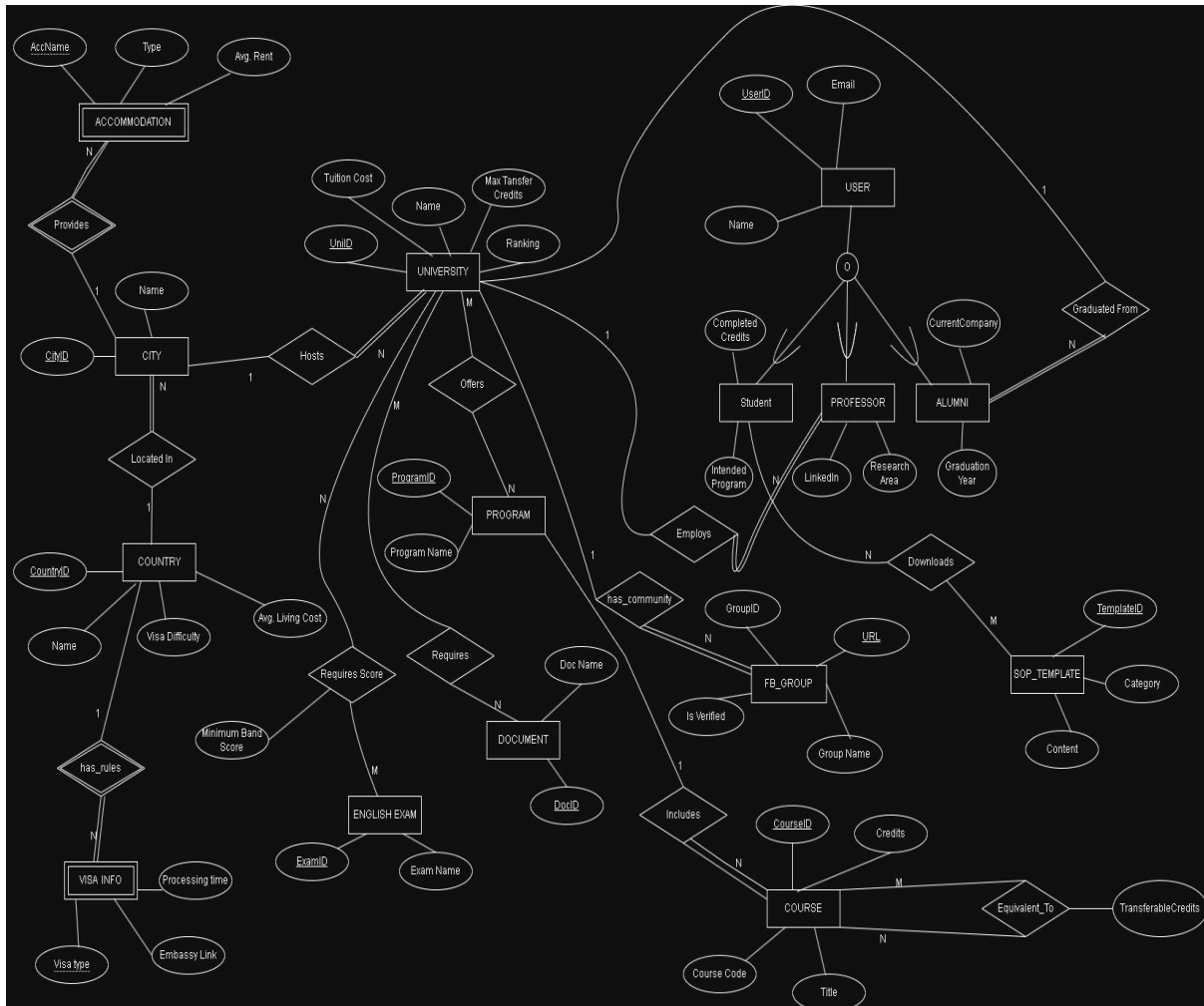
Introduction

The project “Credit Transfer System” is a database platform designed for undergraduate students planning to go abroad. It helps students to explore countries based on Ielts requirements, Tuition fees, visa difficulty and living cost. Users can select their desired academic program and find suitable universities offering those programs. The system allows filtering of universities by ranking cost and credit transfer opportunities. Its core feature is credit equivalency, where students can match their completed courses with transferable credits abroad. It also provides SOP templates, accommodation information. Lastly, the platform includes exam guidance, required documents and visa processing support in one place.

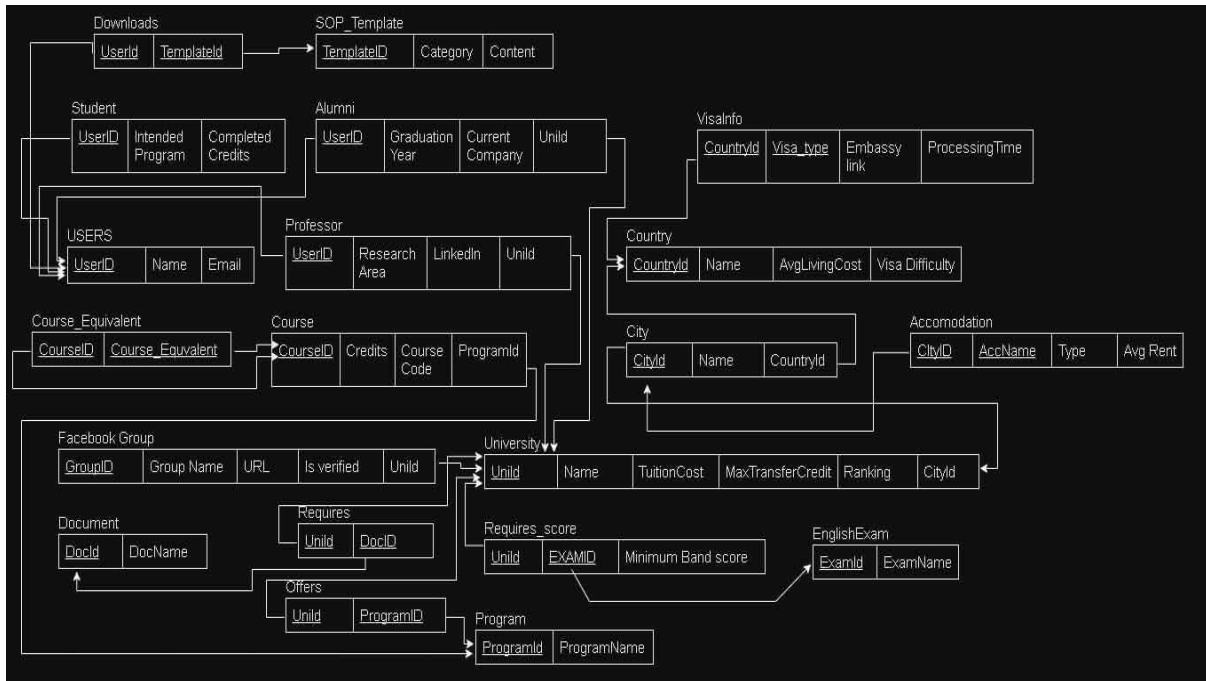
Project Features

ID, Name	Features [3 per member]	
24101411, Taseen Mohammad Faruk	Ft 1	Manage university profiles
	Ft 2	Manage entry requirements
	Ft 3	Manage accommodation options
24101393, Arunima Lodh	Ft 1	Course Directory
	Ft 2	Credit Transfer Calculator
	Ft 3	Statement of Purpose Templates
24101404, Aditya Dey	Ft 1	Manage Programs
	Ft 2	Manage Cities
	Ft 3	Manage Countries

ER/EER Diagram



Schema Diagram



Normalization

- a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.

=> The converted Schema is in 1NF since we followed the basic 8 rules of Schema mapping from the EER Diagram.

- b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?

=> The converted Schema is in 2NF as there are no partial dependencies in the Relational Schema.

- c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

=> Since all the relations are in 2NF, the converted Schema is in 3NF. Also there are no transitive dependencies in the relation.

Frontend Development

Contribution of ID : 24101411, Name : Taseen Mohammad Faruk

- Infrastructure & Security: Built the core database connection (db.php) and the complete user authentication system (login, registration, logout).
- Dashboards: Developed the central hubs for both logged-in students and administrators.
- Complex Data Management: Created the massive University Profile pages and the admin panels to manage accommodations, required documents, and visas.

Contribution of ID : 24101393, Name : Arunima Lodh

- Academic Algorithms: Engineered the complex credit transfer calculator (transfer_checker.php) and the backend mapping tools for course equivalencies.
- Course Directory: Built the public-facing list of university courses and the admin tools to manage them.
- Student Resources: Developed the library of Statement of Purpose (SOP) templates where students can read and save examples.

Contribution of ID : 24101404, Name : Aditya Dey

- Global UI & Search: Managed the global site layout (headers/footers) and built the initial search interface on the landing page (index.php).
- Filtering Engine: Developed the robust backend logic(results.php) that dynamically filters university search results based on user inputs.
- Foundational Admin: Created the administrative interfaces to manage the core lists of academic programs and locations (cities/countries).

Backend Development

Contribution of ID : 24101411, Name : Taseen Mohammad Faruk

- Backend Infrastructure & Security: Engineered the core database connection script (db.php) and developed the complete authentication flow, including secure registration, login, and session-destroying logout functionalities.
- Data Aggregation: Wrote and executed complex, multi-table SQL queries for university_details.php to pull cohesive data regarding English exams, required documents, visas, and housing for specific institutions.
- Requirement & Housing Administration: Built admin_exam_docs.php featuring a sleek, tabbed interface that allows administrators to seamlessly define and link required English exams (like IELTS) and necessary documents to specific universities. Additionally, developed admin_accommodation.php to enable admins to manage and assign housing options (such as dormitories or hostels) to specific cities, tracking average rent costs.
- Dashboards & Smart Entry: Built the secure student dashboard.php and the admin_dashboard.php (featuring live statistical queries). Developed comprehensive data-entry modules for admins(admin_universities.php), including smart logic that auto-generates missing city/country records.

Contribution of ID : 24101393, Name : Arunima Lodh

- Transfer Algorithm Engine: Engineered the mathematical and logical backend for transfer_checker.php , calculating precise credit equivalencies between student inputs and database records.
- Equivalency Mapping System: Built the highly complex backend admin tool (admin_courseequivalent.php) that allows administrators to manually map local courses to host courses and define their transfer weight.
- Database Resource Management: Managed the database connections and queries for the public course directory and the downloadable SOP template library.

Contribution of ID : 24101404, Name : Adittya Dey

- Dynamic Search Engine Backend: Developed the backend logic for results.php , which catches user inputs from the landing page and dynamically generates complex, conditional SQL queries to filter universities.
- Filtering Variables: Implemented the backend logic to filter results by specific parameters like Country, Tuition Cost, Ranking, and Transfer Credits.
- Foundational Database Management: Created the administrative backend queries to add, edit, and delete the core academic programs and geographical locations stored in the database.

Source Code Repository

[Project's Repository Link](#)

Conclusion

In conclusion, our project successfully combines all the designed features and operates as intended. Throughout its development, we worked with several technologies such as CSS, HTML and PHP; gaining practical knowledge along the way. It provided us with hands-on experience in both frontend and backend aspects of development. We also utilized essential database techniques like EER diagrams and Schema Mapping to structure and handle data efficiently. Additionally, the process strengthened our problem-solving abilities and enhanced our teamwork skills. In summary, this project served as a meaningful learning experience, allowing us to convert our concepts into a fully functional and practical system.

References

- 1.Bootstrap: <https://getbootstrap.com/>
- 2.HTML & CSS (MDN): <https://developer.mozilla.org/en-US/docs/Web>
- 3.PHP: <https://www.php.net/>
- 4.MySQL: <https://www.mysql.com/>
- 5.Git: <https://git-scm.com/>
- 6.GitHub: <https://github.com/>